

Saturday Exercises: Population Analysis with Agentic AI

NPC Summer School 2026

Today's exercises: population structure

Same dataset, same corrected fluorescence -- now asking whether cells move together.

Exercise 4 • PCA

Population Structure

113 cells, 4535 frames -- too many dimensions to eyeball. Z-score each cell, then find directions of shared variance across the population.

vs. intuition: no dominant mode -- variance spreads almost linearly across dimensions, a real and informative answer.

Exercise 5 • STA

Spike-Triggered Averaging

Align every cell's trace to one reference cell's spike times and average across events, separating real coupling from noise.

vs. intuition: the reference cell answers cleanly to itself; every other cell stays near its own noise floor.

Time budget: both exercises build directly on Friday's corrected fluorescence -- budget about 2 hours for both.

Exercise 4: Population Structure (PCA)

113 cells is too many dimensions to eyeball -- PCA finds the directions of shared variance.



The problem

113 cells x 4535 frames is too many dimensions to eyeball. Z-score each cell first, or bright cells dominate every component.



Your task

Fit PCA on frames x cells -- the transpose of how the data is stored -- and report how much variance each component explains.



Deliverables

Report variance at PC1, top 3, and top 10; find how many components reach 80-90%; check if PC1's top cells actually move together.



Benchmark

PC1 alone explains 3.2% of variance (top 10: 15.0%); 51 of 113 components are needed for 80-90% -- spread broadly, not concentrated.

Key insight: A near-linear variance curve isn't a failed analysis -- it's a real answer: this population is closer to high-dimensional than low.

PCA: what and why

113 dimensions is too many to eyeball -- PCA finds the directions of shared variance that matter.

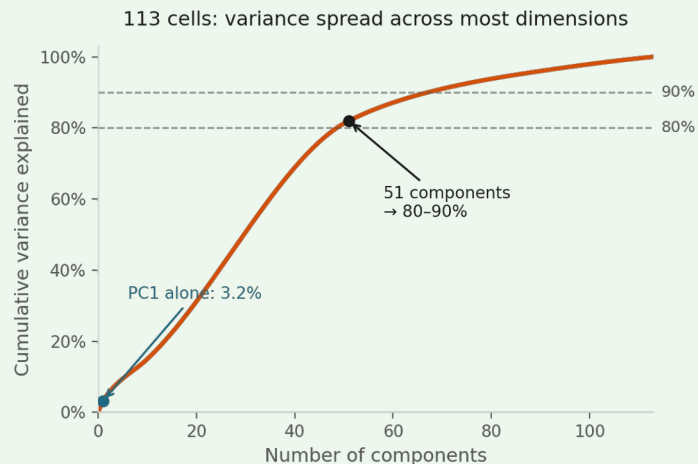
The math

X: 4535 frames x 113 cells, z-scored
PC_k = data projected onto direction k
 $\% \text{ variance}_k = \text{var}(\text{PC}_k) / \text{total variance}$

Z-score each cell, then decompose the frames x cells matrix. Each component v_k is a direction in cell-space; $\% \text{ variance}_k$ is the share of total variance it captures.

Why it matters: a near-linear cumulative-variance curve isn't a failed analysis -- it's a real, informative answer about how this population is organized.

The result



PCA: the geometric picture

Every frame is a point in cell-space -- PCA rotates the axes to point where the data actually spreads.

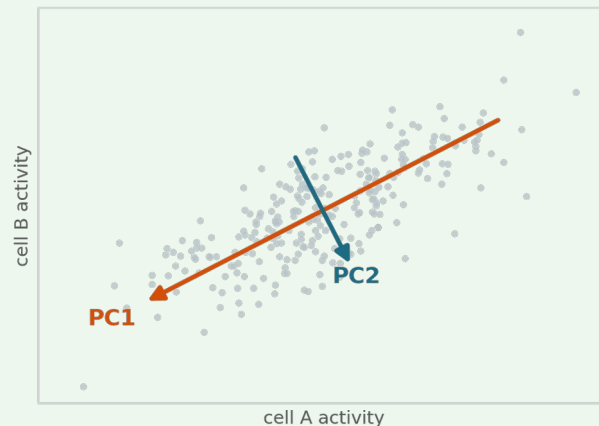
The algebra

v_1 = direction of max spread
 v_2 = next direction, at a right angle to v_1 , and so on

PCA doesn't look at cells one at a time -- it rotates the whole coordinate system so the new axes point wherever the population actually varies most. These axes are exactly the eigenvectors of the cell-by-cell covariance matrix, ranked by eigenvalue.

The picture

Each frame is one point -- PCA finds the long axes



Why it matters: the cell-by-cell axes we started with were arbitrary -- these rotated axes are the ones the data itself prefers.

Exercise 5: Spike-Triggered Average

Does any other cell respond, on average, when one specific cell fires?



The problem

A single trial is dominated by noise. Only averaging across many trigger times separates a repeatable response from that noise.



Your task

Pick a reference cell, get its spike times with `find_peaks`, align every good cell's z-scored trace to those times, and average.



Deliverables

Track standard error per cell per timepoint, then check whether any other cell's average clearly exceeds its own pre-trigger baseline.



Benchmark

Reference cell's own peak response: 0.95. Best other cells: only 0.14-0.16 -- about 6x smaller, indistinguishable from noise.

Key insight: 47 of 112 cells technically exceed their own baseline, but by a trivial margin -- a nominal count without a size check can mislead.

Spike-triggered averaging: what and why

Averaging locks onto a repeatable response and lets unrelated noise cancel out.

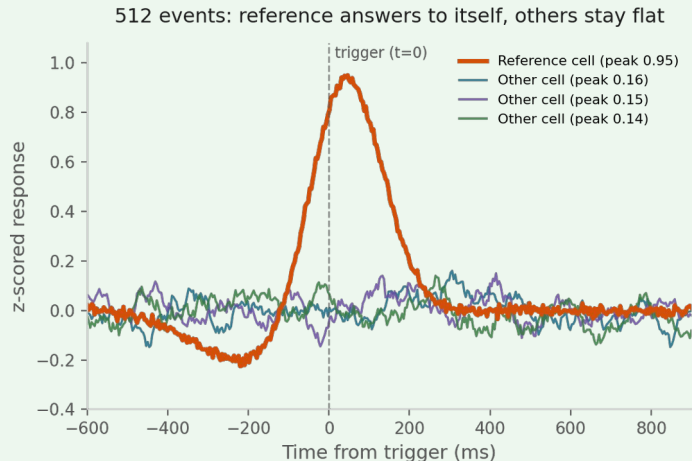
The math

window_i = trace around spike time i
STA = average across all windows
 $\text{SEM} = \text{spread} / \sqrt{(\text{number of spikes})}$

Cut a window of every cell's z-scored trace around each of the reference cell's N spike times, then average. Noise unrelated to the trigger cancels out; a real response survives.

Why it matters: the reference cell answering cleanly to itself is the positive control that proves the method works -- so a flat response elsewhere is a real null, not a failed test.

The result



STA: the geometric picture

Each trial is signal plus independent noise -- stacking and averaging is just vector addition.

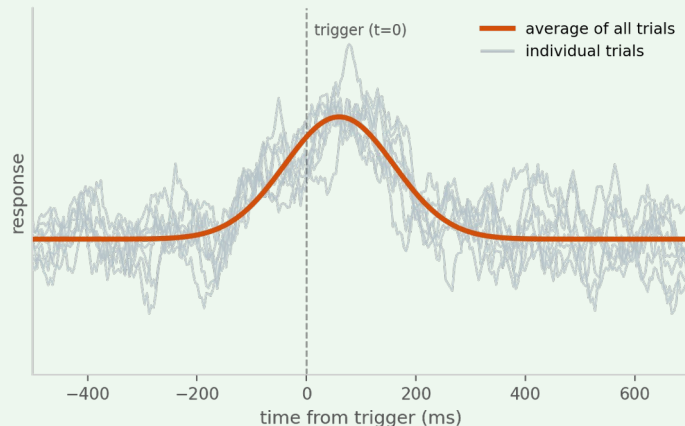
The algebra

$\text{trial}_i = \text{shared signal} + \text{noise}_i$
 $\text{average} = \text{signal} + \text{mean of noise}_i$
independent noise $\rightarrow$ average $\rightarrow 0$

Line up every trial at its spike time and add them like vectors. The shared part adds up every time; independent noise points a different way each trial and washes out. This align-and-average trick is the same idea behind the classic PSTH.

The picture

Stack every trial, average -- noise cancels, signal stays



Why it matters: this is exactly why more trials makes the average cleaner -- it's the law of averages, not a special trick.

Looking back

What these two analyses actually taught you: some structure exists, but it's modest, not dominant.

Exercise 4 • PCA

Population Structure

Variance spread across most of 113 dimensions -- no dominant shared mode. A working positive control (PC1's top cells really do move together) confirms this is real, not noise.

Exercise 5 • STA

Spike-Triggered Averaging

The reference cell answers cleanly to itself -- the method works. No other cell shows more than a trace of coupling: a clean null result, not a failed search.

The common thread: two independent methods agree -- real coordination exists here, but it's small and distributed, not dramatic.